

**Adam/RMSProp  
second moments track**

$$\begin{aligned}\mathbb{E}[g_j^2 \mid e] \\ &= c_e \Sigma_{jj} + 2(\Sigma e)_j^2 \\ &\asymp c_e \Sigma_{jj}\end{aligned}$$

**Spectrum visible**

Aligned or band-limited coordinates  
 $P_t \asymp d_t^{-1/2} (\Sigma + \rho_t I)^{-1/2}$   
 $q_{\text{eff}} \approx 1/2$  (scalar factor ignored)

**Spectrum hidden**

Haar or global Gaussian sketch  
when coordinate variances concentrate  
 $P \asymp cI$   
 $q_{\text{eff}} \approx 0$

**Scaling-law consequences**

visible  $\theta \Rightarrow q_{\text{eff}} = \theta/2$   
 $\Rightarrow K_{\rho, \theta}(n)$   
 $\Rightarrow R(M, N), R_*(C)$   
 $\Rightarrow$  AdamW decay schedule